SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA5108

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 TOTAL MARKS: 50

ANNEXURE A – APPLICATION-BASED QUESTIONS

Student Details

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Code of Conduct

I _____G Dinesh Karthik, 192511085_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G Dinesh Karthik

1. SCENARIO: EVENT COMMUNICATION SECURITY

Question

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Answer

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.

1(a). Reconstruction of the Encrypted Message

The given scenario specifies a fixed shift of 4 positions. This corresponds to a Caesar cipher, a classical substitution technique in which each alphabetic character is replaced by the character located a fixed number of positions away in the alphabet. For this question, the shift is 4 positions forward. Spaces are not shifted and are retained in their original positions.

Alphabet reference used for encryption:

ABCDEFGHIJKLMNOPQRSTUVWXYZ

With a shift of +4, the corresponding mapping is:

A → E B → F C → G D → H E → I F → J G → K H → L I → M J → N K → O L → P M → Q

N → R O → S P → T Q → U R → V S → W T → X U → Y V → Z W → A X → B Y → C Z → D

The given plaintext is:

MEET AT NOON

Each character is shifted four positions forward as follows:

Position	Plaintext	Shift	Ciphertext
1	M	+4	Q
2	E	+4	I
3	E	+4	I
4	T	+4	X
5	A	+4	E
6	T	+4	X
7	N	+4	R
8	O	+4	S
9	O	+4	S
10	N	+4	R

Therefore, the encrypted message is:

QIIX EX RSSR

Explanation of the Transformation

The first word, MEET, is transformed into QIIX. M becomes Q because Q is four positions after M. E becomes I, and the same transformation is applied to the second E. T becomes X. The space between words remains unchanged.

The second word, AT, becomes EX because A shifts to E and T shifts to X. The final word, NOON, becomes RSSR because N shifts to R, O shifts to S, the second O shifts to S, and the final N shifts to R.

Thus, the complete transformation can be represented as: MEET AT NOON → QIIX EX RSSR.

1(b). Retrieval of the Original Message by an Authorized Member

An authorized member who knows the encoding rule can reverse the process. Since encryption shifted every letter four positions forward, decryption shifts every letter four positions backward. The same key value, 4, is therefore used in the reverse direction.

Decryption rule:

Ciphertext character − 4 positions → Plaintext character

The received ciphertext is:

QIIX EX RSSR

The individual characters are reversed as follows:

Ciphertext	Shift Back	Plaintext
Q	−4	M
I	−4	E
I	−4	E
X	−4	T
E	−4	A
X	−4	T
R	−4	N
S	−4	O
S	−4	O
R	−4	N

Therefore:

QIIX EX RSSR → MEET AT NOON

Decryption Procedure

First, the authorized member identifies the fixed shift used by the sender. The assessment specifies a shift of 4 positions. Second, the member takes each alphabetic character of the ciphertext and moves it four positions backward. Third, spaces are retained in their original locations. Finally, the recovered characters are combined to obtain the original message.

For example, Q is four positions after M, so reversing the shift gives M. Similarly, I becomes E and X becomes T. Applying the same reverse operation to every character reconstructs the complete plaintext.

Analysis and Validation

The encryption and decryption results are consistent because applying a +4 shift to the original message produces QIIX EX RSSR, while applying a −4 shift to that ciphertext returns MEET AT

NOON. This provides a direct validation of the reconstructed encrypted message and the reverse process.

The method demonstrates the basic principle of a classical fixed-shift substitution cipher: the same shift is applied to every alphabetic character. Its simplicity makes the process easy for authorized members to perform, but the fixed and limited nature of the shift also makes the method relatively simple to analyze compared with modern encryption systems.

a. Encrypted message using a shift of 4: QIIX EX RSSR

b. The authorized member decrypts by shifting every ciphertext letter 4 positions backward, recovering: MEET AT NOON.

The complete verified transformation is: MEET AT NOON → QIIX EX RSSR → MEET AT NOON.

3. SCENARIO: SECURE MESSAGING SYSTEM

Question

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

a. Demonstrate how the encryption process transforms the message step-by-step.

b. Explain how the repeating keyword influences each character of the ciphertext.

Answer

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

3(a). Step-by-Step Encryption of “HELLO” Using the Keyword “KEY”

The scenario specifies a keyword-based encryption technique. For the given plaintext and keyword, the encryption process is demonstrated using the Vigenère cipher. The Vigenère cipher uses a

keyword to determine the shift applied to each plaintext character instead of using one fixed shift for the complete message.

For calculation, the alphabet is represented using numerical values from 0 to 25, where A = 0, B = 1, C = 2, and so on until Z = 25.

Alphabet-to-number mapping:

A=0, B=1, C=2, D=3, E=4, F=5, G=6, H=7, I=8, J=9, K=10, L=11, M=12

N=13, O=14, P=15, Q=16, R=17, S=18, T=19, U=20, V=21, W=22, X=23, Y=24, Z=25

Given:

Plaintext = HELLO

Keyword = KEY

The plaintext contains five characters, whereas the keyword contains three characters. Therefore, the keyword is repeated until it has the same length as the plaintext.

Repeated keyword:

KEYKE

The aligned plaintext and keyword are:

Position	Plaintext	Key
1	H	K
2	E	E
3	L	Y
4	L	K
5	O	E

The Vigenère encryption formula is:

$$C = (P + K) \bmod 26$$

Here, P represents the numerical value of the plaintext character, K represents the numerical value of the corresponding keyword character, and C represents the numerical value of the resulting ciphertext character. The modulo 26 operation ensures that the result remains within the alphabet range 0–25.

Character-by-Character Encryption

Position 1: H using K

H = 7 and K = 10. Therefore, C = (7 + 10) mod 26 = 17. The number 17 corresponds to R. Thus, H encrypted with K gives R.

Position 2: E using E

E = 4 and E = 4. Therefore, C = (4 + 4) mod 26 = 8. The number 8 corresponds to I. Thus, E encrypted with E gives I.

Position 3: L using Y

L = 11 and Y = 24. Therefore, C = (11 + 24) mod 26 = 35 mod 26 = 9. The number 9 corresponds to J. Thus, L encrypted with Y gives J.

Position 4: L using K

L = 11 and K = 10. Therefore, C = (11 + 10) mod 26 = 21. The number 21 corresponds to V. Thus, L encrypted with K gives V.

Position 5: O using E

O = 14 and E = 4. Therefore, C = (14 + 4) mod 26 = 18. The number 18 corresponds to S. Thus, O encrypted with E gives S.

Complete calculation table:

Pos.	Plain	P	Key	K	Calculation	Cipher
1	H	7	K	10	(7+10) mod 26 = 17	R
2	E	4	E	4	(4+4) mod 26 = 8	I
3	L	11	Y	24	(11+24) mod 26 = 9	J
4	L	11	K	10	(11+10) mod 26 = 21	V
5	O	14	E	4	(14+4) mod 26 = 18	S

Therefore, the encrypted result is:

HELLO + KEY → RIJVS

3(b). Influence of the Repeating Keyword on Each Character

The keyword KEY has only three characters, while HELLO contains five characters. The keyword is therefore repeated from the beginning after its third character. This produces the aligned key sequence KEYKE.

The first plaintext character H is paired with K. The second plaintext character E is paired with E. The third plaintext character L is paired with Y. After reaching the end of the keyword, the sequence starts again, so the fourth plaintext character L is paired with K and the fifth plaintext character O is paired with E.

Plaintext	**H**	**E**	**L**	**L**	**O**
Keyword	K	E	Y	K	E
Ciphertext	R	I	J	V	S

This shows that the keyword does not act as one single fixed shift for the entire message. Instead, different keyword characters provide different shifts at different positions. For example, K corresponds to a shift of 10, E corresponds to a shift of 4, and Y corresponds to a shift of 24 under the A=0 to Z=25 representation.

When the keyword reaches its end, it repeats. Therefore, K is used again at the fourth position and E is used again at the fifth position. The repeated keyword directly determines which shift is applied to each plaintext character.

Overall Encryption Flow

The complete process can be represented as:

Plaintext: H E L L O

Repeated Key: K E Y K E

Numerical P: 7 4 11 11 14

Numerical K: 10 4 24 10 4

Ciphertext: R I J V S

Verification of the Result

The result can be verified by independently checking every character using the same formula C = (P + K) mod 26. The five calculated ciphertext values are 17, 8, 9, 21 and 18, which correspond to R, I, J, V and S respectively. Therefore, the ciphertext RIJVS is consistent with the stated plaintext HELLO and repeated keyword KEY.

The step-by-step calculation also demonstrates that the keyword is not simply appended to the message. Instead, it controls the individual shifts used during encryption. This is the central feature required by the keyword-based encryption scenario in the question.

Analysis and Result

The encryption of HELLO using KEY produces RIJVS. The repeating keyword determines the shift used for each character: K, E, Y, K and E correspond to numerical shifts 10, 4, 24, 10 and 4. Because the keyword is shorter than the message, it repeats to provide a key character for every plaintext character.

The calculation provides a systematic implementation of the keyword-based encryption process and validates the result at every character. The final ciphertext is therefore RIJVS.

a. Using the Vigenère method with plaintext HELLO and keyword KEY, the repeated key is KEYKE and the resulting ciphertext is RIJVS.

b. The keyword repeats after reaching its end, so K, E, Y, K, E are paired with H, E, L, L, O respectively. Each key character determines the shift applied to the corresponding plaintext character.

RUBRICS FOR CALCULATION

The following rubric is reproduced from the supplied assessment document and included as the final page.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30%	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20%	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15%	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10%	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Rubric total weightage: 100% (25% + 30% + 20% + 15% + 10%).